Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. What is the fundamental matrix? **1 / 1 point**

- ○ The matrix that represents the intrinsic properties of a camera
- ◉ The matrix that describes the relative geometry between two cameras
- ○ The matrix that encodes the colors of an image
- ○ The matrix that represents the intensity values of an image

Correct
The fundamental matrix in stereo vision describes the relative geometric relationship between two camera views. It encodes essential information about how corresponding points in one camera's image relate to points in the other camera's image. By knowing the fundamental matrix, you can determine the epipolar geometry.

2. What is the relationship between the essential matrix and the fundamental matrix? **1 / 1 point**

- ◉ The essential matrix explains 3D information, while the fundamental matrix explains 2D information.
- ○ The essential matrix and the fundamental matrix are the same.
- ○ The essential matrix is a subset of the fundamental matrix.
- ○ There is no relationship between the two.

Correct
The essential matrix (E) explains 3D information, representing the relative translation and rotation between two cameras. The fundamental matrix (F) explains 2D information, describing how corresponding points in one image are related to the other image through epipolar lines. The relationship between E and F is E = K^T * F * K, where K is the camera calibration matrix. E is used for 3D reconstruction, while F is used for 2D correspondence.

3. Which method can be used to extract the rotation and translation matrix from the essential matrix? **1 / 1 point**

- ○ Principal Component Analysis (PCA)
- ◉ Singular Value Decomposition (SVD)
- ○ Fast Fourier Transform (FFT)
- ○ Gradient Descent

Correct
To extract rotation and translation matrices from an essential matrix, you typically use Singular Value Decomposition (SVD). Singular Value Decomposition is a mathematical technique that can be used to decompose the essential matrix into its constituent parts, which includes the rotation and translation matrices.